

Primary sources & further reading

Part 03 — Natural Language Processing · 03-nlp/papers/PAPERS.md · 03-nlp/resources/RESOURCES.md



Papers — NLP & Transformers

`bash ../_tools/fetch_papers.sh 03-nlp` downloads these into this folder.

Read #1 twice. Then read it again in a month.

#	Paper	Year	Why	Link
1	Attention Is All You Need — Vaswani et al.	2017	The transformer. Eight pages that redirected the field. Read alongside <i>The Annotated Transformer</i> .	arXiv:1706.03762
2	BERT — Devlin et al.	2018	Bidirectional pretraining + fine-tuning; created the "pretrain then adapt" era	arXiv:1810.04805
3	Improving Language Understanding by Generative Pre-Training (GPT-1) — Radford et al.	2018	The decoder-only bet	PDF
4	Neural Machine Translation by Jointly Learning to Align and Translate — Bahdanau et al.	2014	Where attention was actually invented	arXiv:1409.0473

Tokenization

Paper	Takeaway	Link
Neural Machine Translation of Rare Words with Subword Units (BPE) — Sennrich et al., 2015	BPE for NLP	arXiv:1508.07909
SentencePiece — Kudo & Richardson, 2018	Language-agnostic, no whitespace pre-tokenization	arXiv:1808.06226
Subword Regularization — Kudo, 2018	The unigram LM tokenizer	arXiv:1804.10959

Architecture & efficiency

Paper	Takeaway	Link
RoFormer (RoPE) — Su et al., 2021	Rotary position embeddings; in almost every modern LLM	arXiv:2104.09864
Train Short, Test Long (ALiBi) — Press et al., 2021	Length extrapolation via attention bias	arXiv:2108.12409
Fast Transformer Decoding: One Write-Head (MQA) — Shazeer, 2019	The KV-cache problem, stated early	arXiv:1911.02150
GQA: Training Generalized Multi-Query Transformer Checkpoints — Ainslie et al., 2023	The modern default	arXiv:2305.13245
FlashAttention — Dao et al., 2022	IO-aware exact attention	arXiv:2205.14135
FlashAttention-2 — Dao, 2023	Better parallelism and work partitioning	arXiv:2307.08691
GLU Variants Improve Transformer (SwiGLU) — Shazeer, 2020	The modern FFN	arXiv:2002.05202
Root Mean Square Layer Normalization (RMSNorm) — Zhang & Sennrich, 2019	Cheaper norm, same result	arXiv:1910.07467
On Layer Normalization in the Transformer Architecture — Xiong et al., 2020	Why Pre-LN beat Post-LN	arXiv:2002.04745
T5: Exploring the Limits of Transfer Learning — Raffel et al., 2019	Everything as text-to-text; a huge ablation study	arXiv:1910.10683
Efficient Transformers: A Survey — Tay et al., 2020	Map of the efficiency landscape	arXiv:2009.06732
Mamba: Linear-Time Sequence Modeling — Gu & Dao, 2023	The serious non-attention challenger	arXiv:2312.00752

Embeddings & retrieval (bridges to Module 08)

Paper	Takeaway	Link
Efficient Estimation of Word Representations (word2vec) — Mikolov et al., 2013	Meaning as geometry	arXiv:1301.3781
Sentence-BERT — Reimers & Gurevych, 2019	Usable sentence embeddings via siamese training	arXiv:1908.10084
Dense Passage Retrieval (DPR) — Karpukhin et al., 2020	Dense retrieval beats BM25	arXiv:2004.04906
Text Embeddings by Weakly-Supervised Contrastive Pre-training (E5) — Wang et al., 2022	Modern embedding recipe; note the instruction prefixes	arXiv:2212.03533

The best explainers ever written about transformers

- **The Illustrated Transformer** — Jay Alammar: <https://jalammar.github.io/illustrated-transformer/>

- **The Annotated Transformer** — Harvard NLP (paper + line-by-line code): <https://nlp.seas.harvard.edu/annotated-transformer/>
- **Let's build GPT from scratch** — Karpathy (2h video, builds it live): <https://www.youtube.com/watch?v=kCc8FmEb1nY>
- **3Blue1Brown — Attention in transformers, visually**: <https://www.3blue1brown.com/topics/neural-networks>
- **Transformer Circuits / induction heads** — Anthropic: <https://transformer-circuits.pub/>

Resources — NLP & Transformers

Start here

Resource	Why	Link
Karpathy — Let's build GPT from scratch	2 hours, builds a working transformer live. The best single artifact in this module.	https://www.youtube.com/watch?v=kCc8FmEb1nY
The Illustrated Transformer — Jay Alammar	The diagrams everyone else copies	https://jalammar.github.io/illustrated-transformer/
The Annotated Transformer — Harvard NLP	The paper with runnable code beside each paragraph	https://nlp.seas.harvard.edu/annotated-transformer/
Hugging Face NLP Course (free)	Practical, current, hands-on	https://huggingface.co/learn/nlp-course
Stanford CS224N — NLP with Deep Learning	The university treatment; lectures on YouTube	https://web.stanford.edu/class/cs224n/
Speech and Language Processing — Jurafsky & Martin (free draft)	The field's textbook, updated yearly	https://web.stanford.edu/~jurafsky/slp3/
3Blue1Brown — Transformers & attention	Visual intuition for the matrix operations	https://www.3blue1brown.com/topics/neural-networks

Repositories

Repo	What's inside
https://github.com/karpathy/nanoGPT	~300 lines of readable GPT. Train one on your laptop.
https://github.com/karpathy/minGPT	The even smaller pedagogical version
https://github.com/karpathy/minbpe	BPE in a few hundred clean lines — compare with your own
https://github.com/huggingface/transformers	The library; read <code>modeling_llama.py</code> once, properly
https://github.com/huggingface/tokenizers	Fast Rust tokenizers with a Python API
https://github.com/openai/tiktoken	OpenAI's BPE tokenizer
https://github.com/google/sentencepiece	The tokenizer behind T5, Llama and many others
https://github.com/Dao-AI-Lab/flash-attention	The reference implementation
https://github.com/UKPLab/sentence-transformers	Sentence embeddings — the on-ramp to Module 08
https://github.com/explosion/spaCy	Industrial classical NLP: NER, parsing, pipelines
https://github.com/bentrevett/pytorch-seq2seq	Clear tutorials from RNN seq2seq up to transformers

Clone the starter set with: `bash ../_tools/clone_repos.sh 03-nlp`

Interactive / visual tools

- **Transformer Explainer** (interactive GPT-2 in your browser) — <https://poloclub.github.io/transformer-explainer/>
- **BertViz** (attention visualisation) — <https://github.com/jessevig/bertviz>
- **Tiktokenizer** (see tokens live) — <https://tiktokenizer.vercel.app/>
- **Hugging Face tokenizer playground** — <https://huggingface.co/spaces/Xenova/the-tokenizer-playground>
- **Transformer Circuits** (mechanistic interpretability) — <https://transformer-circuits.pub/>

Datasets & leaderboards

- Hugging Face Datasets — <https://huggingface.co/datasets>
- GLUE / SuperGLUE — <https://gluebenchmark.com/> · <https://super.gluebenchmark.com/>
- MTEB (embedding leaderboard — read this before choosing an embedding model) — <https://huggingface.co/spaces/mteb/leaderboard>
- Papers with Code — NLP — <https://paperswithcode.com/area/natural-language-processing>